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DEPARTMENT OF COMPUTER SCIENCE AND TECHNOLOGY,
BIOENGINEERING, ROBOTICS AND SYSTEM ENGINEERING

SOCIAL ROBOTICS

Second Assignment

Perceived Intelligence Experiment

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1 Introduction

Social robotics is an interdisciplinary field that integrates robotics, artificial intelligence, and human-computer interaction to create robots capable of engaging with people in meaningful and natural ways [1].

These robots, often designed with humanoid characteristics, are increasingly being employed across various domains, such as healthcare, education, retail, and hospitality, where social interactions are central to the experience, and one of the most iconic social robots,Pepper, developed by SoftBank Robotics [3].

In this project, we harness Pepper's social and interactive potential as assistant in a restaurant. For this a programming environment specifically designed for Pepper called Choregraph is used, in which we implement features such as dialog-based interactions, timelines for event sequencing, and switch case mechanism to handle dynamic user scenarios.

2 Perceived Intelligence Experiment

2.1 Choice of Scenario

2.1.1 Part1

Consists of two subparts (scenarios)

- Scenario 1: Eat Here

  This scenario highlights Pepper's ability to perform complex tasks like guiding users to a location and presenting a menu. It reflects real-world situations in hospitality, and showing Pepper's intelligence, and adaptability. Actions like physical movement and empathetic gestures make the interaction engaging and relatable, reinforcing Pepper's role as a helpful assistant.
- Scenario 2: Take away The takeaway scenario showcases Pepper's efficiency in handling quick, task-oriented interactions. It focuses on maintaining emotional intelligence through empathetic gestures and verbal cues, ensuring Pepper is perceived as both functional and personable.

2.1.2 Part2

This scenario highlights Pepper's ability to adapt to user's choice and responding with tailored gestures and dialog, Pepper demonstrates its ability to provide personalized assistance, this showcases also its emotional intelligence and anthropomorphic qualities.

2.2 Experiment design and components in Choregraphe

2.2.1 Part1

The design consisted of

- **Set Language:** Ensures that Pepper communicates in the appropriate language for the user.
- **Greeting Client:** Initiates the interaction with a polite, welcoming phrase accompanied by a friendly gesture.
- **Assistance Module:** Pepper offers assistance by asking about the user's needs, demonstrating its proactive and service-oriented behavior.

- **Empathy Module:** Displays empathetic gestures or expressions to connect emotionally with the user and create a more human-like interaction.
- **Speech Recognition:** Processes user input by listening for keywords like *here* or *take*, guiding subsequent actions based on their choice.
- **Switch Case:** Routes the flow based on user input:
  - **Here:** Leads to the *Eat here!* sequence.
  - **Take:** Initiates the *Take away!* sequence.
- **Eat Here!:**
  - Pepper confirms the choice and invites the user to follow it.
  - **Follow Me:** Pepper guides the user to a predefined location, enhancing its interactive and functional abilities.
  - **Move To:** The robot physically shows to the designated location.
  - **Show Menu:** Shows where exactly is the menu.
  - **Animated Say:** Ends with a cheerful message, such as *Enjoy your meal!*, to conclude the interaction.
- **Take Away!:**
  - Confirms the takeaway choice with appropriate verbal and physical cues.
  - Maintains Pepper's empathetic and service-oriented behavior even in brief interactions.

2.2.2 Part2

This part shares approximately the same components, it extends the interaction by introducing a new decision point based on the user's choice of ”dinner” or ”lunch.” The Switch Case directs the flow to the respective sequence. Each branch incorporates additional empathetic expressions tailored to the selected option. For instance, in the ”dinner” branch, Pepper acknowledges the choice with relevant verbal responses and empathetic gestures specific to a dinner scenario, while the ”lunch” branch mirrors this interaction with appropriate adjustments. This customization enhances the interaction's relevance and perceived intelligence.

3 Questionnaire

3.1 Questionnaire structure

The questionnaire used for this experiment is composed of two parts. The first part aims to define the age group and gender of the respondents, as well as how they generally perceive Pepper's assistance and to what level they trust it. The second part involves implementing the *Goodspeed scale* questionnaire. The *Goodspeed scale* questionnaire is a tool used to evaluate the user perception of the robot's capabilities and characteristics [2]. It is used in various aspects of human-robot interaction. It consists of five key dimensions, each measured using a series of semantic scales from 1 (strongly disagree) to 5 (strongly agree). The five key dimensions are the following:

- **Anthropomorphism:** Evaluate how human-like the robot behaves.
- **Animacy:** Evaluate the robot's lifelikeness level.
- **Likeability:** Evaluate the level of positivity the user feels toward the robot.
- **Perceived Intelligence:** Evaluate the robot's perceived intelligence and abilities.
- **Perceived Safety:** Evaluates whether the robot is seen as safe or not.

Section	Items		
Anthropomorphism	Fake	-	Natural
	Machinelike	-	Humanlike
	Unconscious	-	Conscious
	Artificial	-	Lifelike
	Moving rigidly	-	Moving elegant
Animacy	Dead	-	Alive
	Stagnant	-	Lively
	Mechanical	-	Organic
	Artificial	-	Lifelike
	Inert	-	Interactive
	Apathetic	-	Responsive
Likeability	Dislike	-	Like
	Unfriendly	-	Friendly
	Unkind	-	Kind
	Unpleasant	-	Pleasant
	Awful	-	Nice
Perceived Intelligence	Incompetent	-	Competent
	Ignorant	-	Knowledgeable
	Irresponsible	-	Responsible
	Unintelligent	-	Intelligent
	Foolish	-	Sensible
Perceived Safety	Anxious	-	Relaxed
	Agitated	-	Calm
	Quiescent	-	Surprised

Figure 1: The Godspeed Questionnaire Series [2]

For the second part of the questionnaire, we only tested anthropomorphism, animacy, likeability, and perceived Intelligence.

3.2 Questionnaire result

After conducting the questionnaire on 25 respondents, we obtained the following results:

3.2.1 Age group

Over 23 respondents were in the age gap between 18 and 30 years old, while only 2 were above 30 years old (see Figure 2).

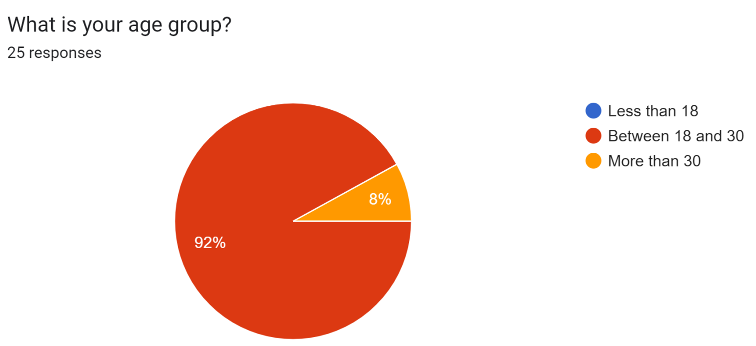

Figure 2: Age group

3.2.2 Gender group

Over 13 respondents were male and 12 were female (see Figure 3).

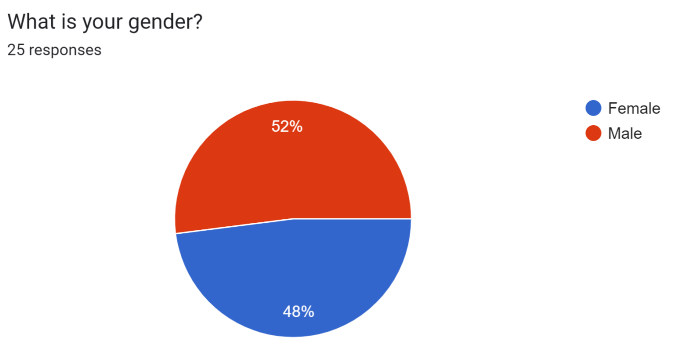

Figure 3: Gender group

3.2.3 Information clearness

Most of the respondents find the information provided by Pepper very clear and understandable (see Figure 4).

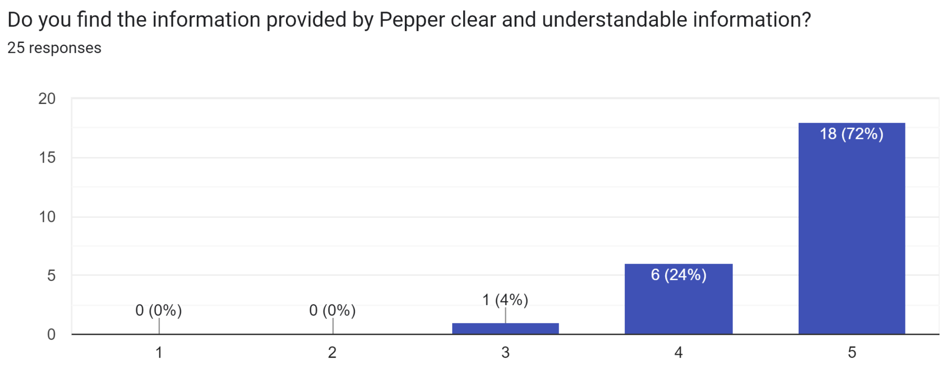

Figure 4: Information clearness

The vast majority of the respondents find Pepper to be helpful in assisting the client at the restaurant (see Figure 5).

3.2.4 Client assistance

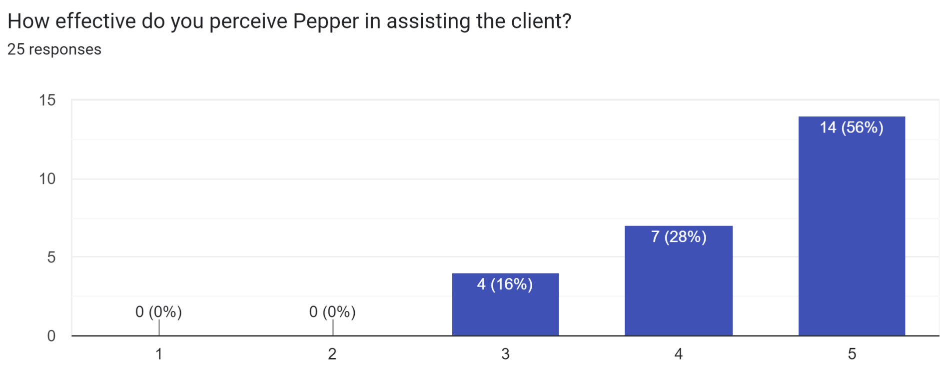

Figure 5: Client assistance

3.2.5 Level of comfort when dealing with Pepper

Most respondents find themselves comfortable with Pepper for a similar application, while only 2 respondents are not (see Figure 6).

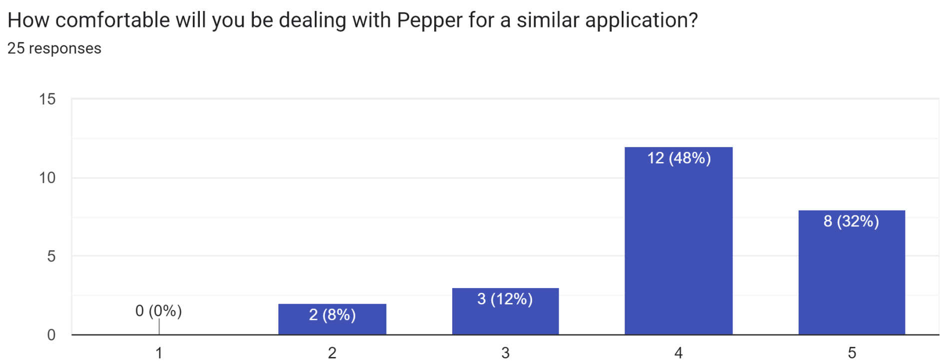

Figure 6: Level of comfort when dealing with Pepper

3.2.6 Comparing the perceptions of Pepper's assistance to that of a human assistant

The results for this part are quite variant, over 16% of the respondents, do not find Pepper that efficient in assisting clients comparing to human assistance, while over 64% perceive it to be useful (see Figure 7).

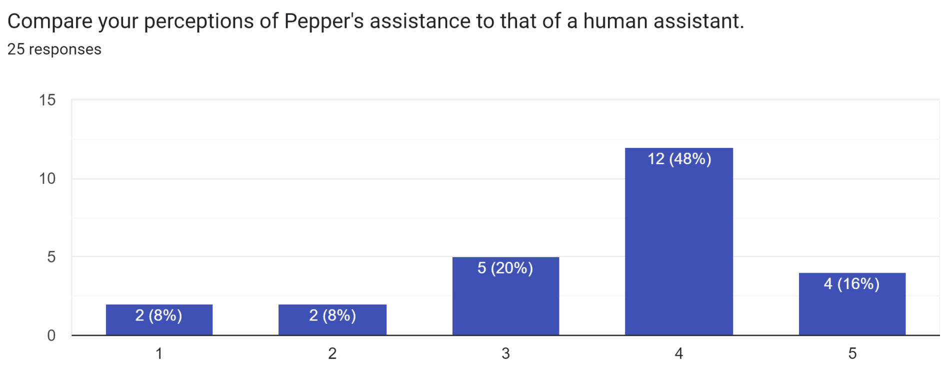

Figure 7: Comparing the perceptions of Pepper's assistance to that of a human assistant

3.2.7 Supporting Pepper as a restaurant assistant

The majority of the respondents agree on supporting Pepper as a restaurant assistant (see Figure 8).

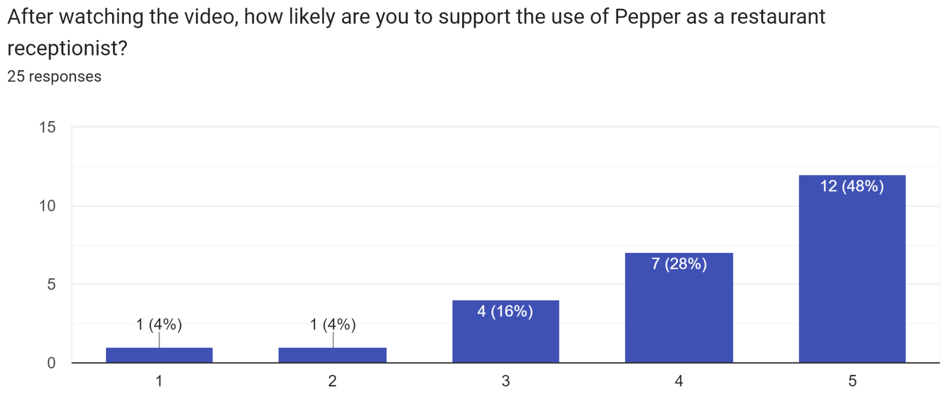

Figure 8: Supporting Pepper as a restaurant assistant

3.2.8 Goodspeed scale questionnaire

Anthropomorphism: Pepper was perceived as moderately natural and humanlike but still retained characteristics of being artificial and slightly machinelike (see Figure 9).

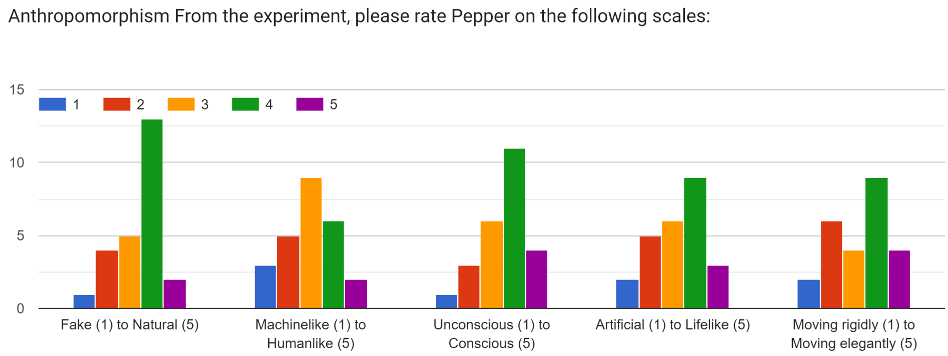

Figure 9: Anthropomorphism

Animacy: Participants generally rated Pepper as more alive, lively, and interactive, but still somewhat mechanical and less organic (see Figure 10).

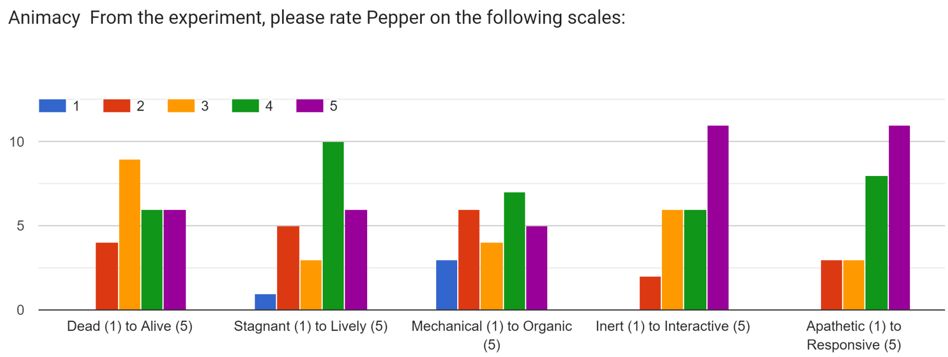

Figure 10: Animacy

Likeability: Participants found Pepper to be generally friendly, kind, and pleasant, with overall positive ratings for likeability (see Figure11).

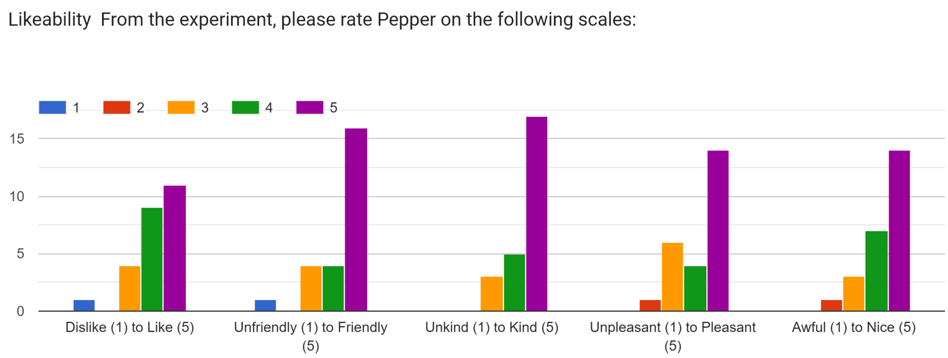

Figure 11: Likeability

Perceived Intelligence: Pepper was rated as relatively competent, knowledgeable, and intelligent, with a perception of being somewhat responsible and sensible (see Figure 12).

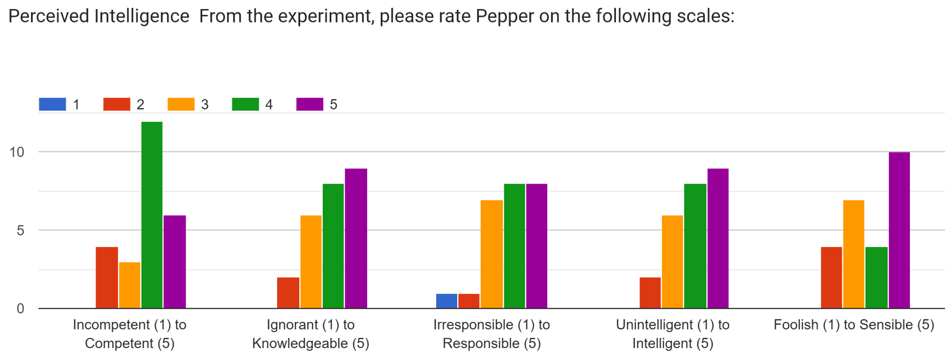

Figure 12: Perceived Intelligence

4 Survey analysis

4.1 T-Test Analysis

We conducted a T-test analysis by comparing responses to two specific survey questions:

- The first question asked participants to rate their overall impression of Pepper's intelligence on a scale from "Not intelligent (1)" to "Intelligent (5)".
- The second question asked participants to evaluate Pepper's decision-making ability, ranging from "Foolish (1)" to "Sensible (5)". This comparative analysis aimed to determine if there were statistically significant differences in the perception of Pepper's intelligence in these different contexts.

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4.2 Hypotheses Formulation

Null Hypothesis (H0): There is no statistically significant difference in the perception of Pepper's intelligence between the overall assessment of intelligence and the evaluation of its decision-making.

Alternative Hypothesis (H1): There is a statistically significant difference in the perception of Pepper's intelligence between the two evaluations.

4.3 Results of the T-test

The T-test analysis produced the following results:

- **T-statistic:** 1.16
- **P-value:** 0.256

These results indicate that there is no statistically significant difference in the average ratings for these two aspects of perceived intelligence.

From these results, we maintain the null hypothesis (H0), suggesting that there are no significant differences in the perception of Pepper's intelligence between the two evaluated contexts.

This result implies that participants consistently thought Pepper was intelligent, regardless of the situation. It indicates that participants generally perceive Pepper's intelligence at a similar level, both in terms of their overall impression and its decision-making ability. This constancy emphasizes the significance of sustaining a constant degree of perceived intelligence throughout various interactions, which is essential for the development and use of social robots like Pepper.

5 General conclusion

The project used a user questionnaire and interactive situations to evaluate Pepper's perceived intelligence as a restaurant assistant. Pepper's position in friendliness was supported by the participants' perceptions of it as being clear, helpful, and easy to interact with. It is effective, as evidenced by positive likeability, animacy, and perceived intelligence scores; yet, anthropomorphic qualities are still moderate.

6 Work devision

The work for this assignment was divided as follows:

- **Model implementation in Choregraphe:** Aicha and Roumaissa.
- **Testing on the Pepper robot:** Aicha and Roumaissa.
- **Making the questionnaire:** Aicha.
- **Editing and uploading the video:** Roumaissa.
- **Report writing:** Introduction and Perceived Intelligence parts: Roumaissa. Questionnaire part: Aicha. Survey analysis: Aicha and Roumaissa.